

ASPHALT BATCH PLANT

Basic specifications and construction (Standard Equipment)

The **ASPHALT BATCH PLANT** is an asphalt mixing plant used for producing Hot Mix Asphalt (HMA), which is used to pave the road. Asphalt is made of bitumen, aggregate and filler, sometimes collectively known as blacktop or asphalt concrete.





Note: Pictures are visual reference only; not intended request specific brand

MAIN COMPONENTS

- Cold aggregate supply system
- Drying drum
- Burner
- Hot aggregate elevator
- Dust collector
- Vibrating screen
- Hot aggregate storage hopper
- Weighing and mixing system
- Filler supplying system
- Finished asphalts storage silo
- Bitumen supplying system
- All electrical, hydraulic, water lines shall be supplied to make system operational less liquid bitumen asphalt product
- All connection points shall be clearly labelled identifying their purpose.

MOBILE ASPHALT PLANT

- Mobile/Portable Asphalt Plant with transport frame chassis which can which is portable type asphalt concrete plant, features modular structure design and transport frame chassis, fast and easy installation
 - **Output capacity range 140-160 Metric Tons ton per hour**
 - **RAP mixing Capable 30% - 50%**
- **Chassis**
 - Chassis One (1): Four independent feed hoppers, dryer and automatic burner mounted on one chassis;
 - Chassis Two (2): Carries the vertical elevator, screening, weighing and mixing tower, control cabin, primary cyclones and secondary bag filter
- Installation and Self erection of the main components achievable within four (4) hours and operational with the addition of a bitumen tank and reclaimed filler silo

MAIN COMPONENTS (minimums)

- Main hopper capacity (heaped): minimum 5.0 yd³
- Hopper loading width: 3.0 yd
- Belt feeder centers: 4 feet
- Belt width: 1.5 feet
- Belt feeder drive: 2.2 kW
- Vibrator fitted to sand hopper
- Collecting conveyor width: 1.5 feet
- Collecting Conveyor drive: 2.2 kW
- Drum
- Bins (7 total)
- Asphalt Cement tank 25,000 gallon
- Scales
- Fuel pump
- Control station/house / Operation controls

HOT STONE ELEVATOR, SCREEN & MIXING SECTION

- Elevator capacity: 90 t/h
- Elevator drive: 7.5 kW
- Screen decks: 2 - (4 size + rejects)
- Screen length: 3.0 yd
- Screen width: 1.5 yd
- Screen drive: 7.5 kW
- Hot storage capacity: 50 tons
- Hot storage bins: 4
- Aggregate weigh hopper capacity: 2200 lbs
- Bitumen weigh hopper capacity: 450 lbs
- Filler weigh hopper capacity: 450 lbs
- Paddle mixer capacity: 2200 lbs
- Paddle mixer drive: 2 x 11.0 kW

AGGREGATE DRYER/MIXER DRUM (counter flow)

- Dryer drum diameter: 1.5 yd
- Dryer drum length: 7 yd
- Dryer drive: 22.0 kW
- Dryer capacity: 80 t/h

BINS

- Aggregate (4 each): virgin aggregate, crushed stone, sand, and gravel
- RAP Reclaimed Asphalt Pavement (3 each)
- Scalping screen between cold feed bins and mixing drum

AIR VOLUME

- Dryer: 23,500 – 27,500 yd³/h
- Mixing section: 6300 - 6700 yd³/h
- TOTAL: 29,800 – 34,200

BITUMEN/ASPHALT CEMENT STORAGE TANK

- Capacities from 25,000 Gallon
- Two/three axle chassis depending on capacity
- High density insulation and robust external finish
- Thermostatically controlled system
- Self-contained heating system to maintain product temperature
- Integral pump permitting loading and discharge of product
- Easy to operate and requires minimal maintenance

HEATING SYSTEM

- Fully automatic high efficiency burners
- Minimum 500-gallon burner fuel tank
- Stainless steel flame liners
- Thermal fuse protected fuel line

WEIGHT SCALE FOR DISPENSED MATERIAL

- Truck Scale and Silo/chute scale

PUMPING SYSTEM

- Integral pump
- Metering system on AC pump
- Auxiliary hydraulically driven
- Minimum 7400 – 8200 gallon per hour capacity

CONTROL STATION

- Operator's control station equipped to operate and monitor all functions of Mobile Asphalt Plant and Bitumen tank
- Station equipped with Heating, Ventilation, Air Conditioning (HVAC)
- Equipped with 125 VAC power
- Control panels for all functions of mobile asphalt plant
- All controls clearly labeled, illuminated
- High quality interior and exterior lights

SAFETY

- Optimum night lighting
 - **A minimum of (12 each) – 2000 lumen each quartz halogen flood lights with switch to Red light night operations with integrated wiring harnesses mounted to mixer**
 - Withstand up to 15.3 Grms vibration at 24-2000Hz / 4.65Grms at 12-2000Hz
 - Stop, Turn, Tail, Warning lights (rear)
 - Clearance and side marker lights
- Minimum of 8 integrated tie down locations on each side as determined by ATTILA certification
- Equipped with sufficient lifting points by use of crane; clearly labeled **Lift Point**
- All steps and ladders made of slip resistant material
 - Handrails for accessing work platforms made of slip resistant material

PAINT/CORROSION PROTECTION

- Manufacturer standard commercial color
- Maximum manufacturer's factory rustproof/corrosion protection package throughout
 - Minimum **5 year** life expectant on rustproof/corrosion protection
 - Automotive grade primer/paint
 - Military grade rustproofing throughout machine
 - Shall comply with Military Specifications: MIL-PRF-16173, Grade 4 and MIL-C-83933
 - Electrical/hydraulic lines covered/protected from damage during machine operation
- Machine, including all parts and accessories, shall be constructed and finished thoroughly without blemish, flaws, defects and/or sharp edges/corners.
- Welding and post heat treatment procedures shall be performed in accordance with American Welding Society (AWS) Welding Code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings
- Machine shall be thoroughly cleaned. Loose, spattered, or excess welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly

WARRANTY

- 5 Year Warranty Plan

TRAINING

- Conduct initial set-up, tear-down, and operating training.
- Conduct yearly refresher training for a minimum of 3 personnel.

PARTS and SERVICE MANUALS

- Complete set of manuals available at time of delivery to the Air Force, **No Exceptions**
- Includes parts breakdown, service manual and scheduled lubrication chart, repair processes for any and all parts/components of machine, **No Exceptions**
- Two (2) complete sets of Operators, Parts and Service manuals to include drive engine, auxiliary engine (if equipped), transmission, and any/all mounted and/or attachments shipped with machine
- Provide diagnostic troubleshooting software/devices to end users
 - Air Force vehicle maintenance personnel shall have capability to diagnose, troubleshoot, and service machine locally

- Diagnostics capable for all components of machine, **No Exceptions**
 - All software and firmware shall function as intended, and shall be free of vulnerabilities (i.e. defects, bugs, malware, etc.), either intentionally or unintentionally inserted as part of the software/firmware throughout the lifecycle.
 - Any Windows-based software shall be compatible with Windows 10 (64-bit) operating system.

CALIBRATION AND MAINTENANCE

- Within calendar year / Contractor can coordinate with user upon award
- Contractor must confirm the maintenance schedule and dates with the Government prior to equipment delivery
- Vendor provides **24 month** warranty service for any/all components of machine regardless of manufacturer of prime mover, accessory, and/or attachments as delivered to the Air Force
- Bi-annual Basic Maintenance
- Software updates
- Standard Wear-Part replacement plan
- Annual inspection
- 24/7 technical support option
- Spare parts list with part numbers

MANUFACTURER/DEALERSHIP will include

- Initial operator's familiarization set-up (minimum **40 hours** – on site)
- Initial maintenance familiarization set-up (minimum **40 hours** – on site)
- Military Marking / Data Plate(s)
- Data plates will include the following information, at a minimum

NOMENCLATURE: _____

MAKE AND MODEL: _____

MFR'S SERIAL #: (VIN _____)

REGISTRATION #: _____

NSN: _____

VEHICLE CURB WEIGHT: _____ LBS

PAYLOAD, MAXIMUM: _____ LBS

GROSS WEIGHT, MAX: _____ LBS

GROSS WEIGHT COMB MAX: _____ LBS

DATE OF DELIVERY: _____

WARRANTY: _____ MO _____ MILES

CONTRACT #: _____

SHIP TO: _____

MARK FOR: _____

US PROPERTY